Examining the Impact of Blend Ratio and Annealing Time on the Densities of Thin Squaraine Films

Dr. Kristen Burson
Dr. Christopher Collison
Tyler Wiegand

Hudson Smith '21

Acknowledgements

Special thanks to:

Dr. Kristen Burson for all of her help as my mentor for this project.

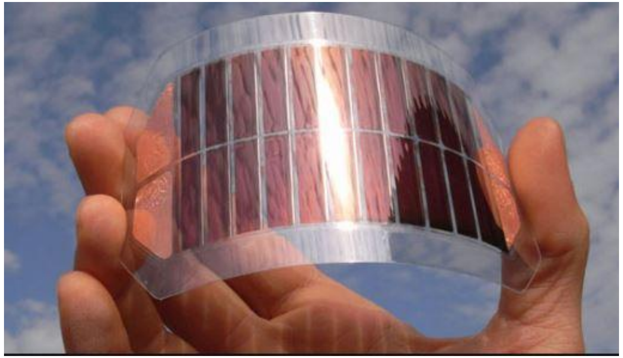
Marvin Lopez ‘23 for his help with data collection and analysis.

Dr. Christopher Collison and Tyler Wiegand for their help in constructing and analyzing these samples.

Bentley Wilking ‘21 and Sarah Gyurina ‘22 for their help with collection of photographic images.

My parents, Caroline Cahill and William Smith for their constant support.

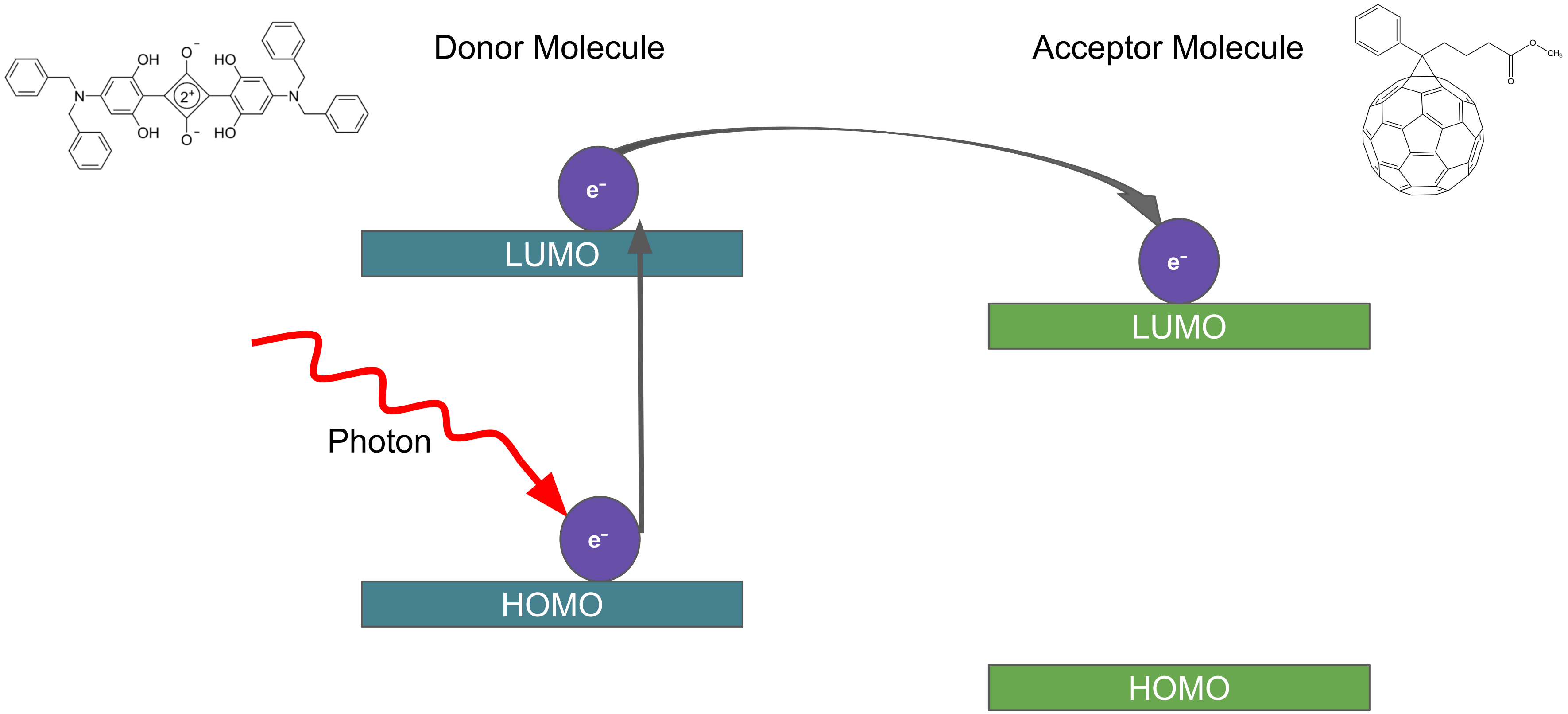
Why Organic Photovoltaics?



Organic Photovoltaics are different from traditional solar cells because they are

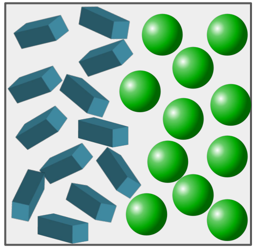
1. Flexible
2. Semi-transparent
3. Lightweight

Technology: The Bulk Heterojunction

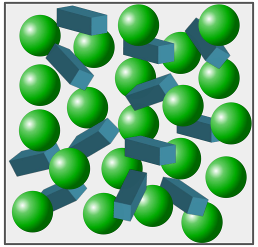


Morphology and OPV Efficiency

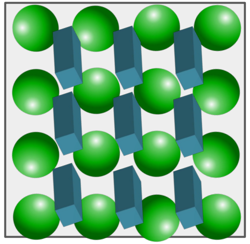
- As crystalline structures form, the efficiency of squaraine-fullerene OPV devices decreases
- This is because intermixing limits charge transfer from donor to acceptor
  - Charge transfer energy has a direct correlation with Open Circuit voltage of devices
- Aggregation is also necessary to some extent in squaraine-fullerene OPVs as it broadens the spectra of the squaraine donor.



no electron transfer between donor and acceptor molecules

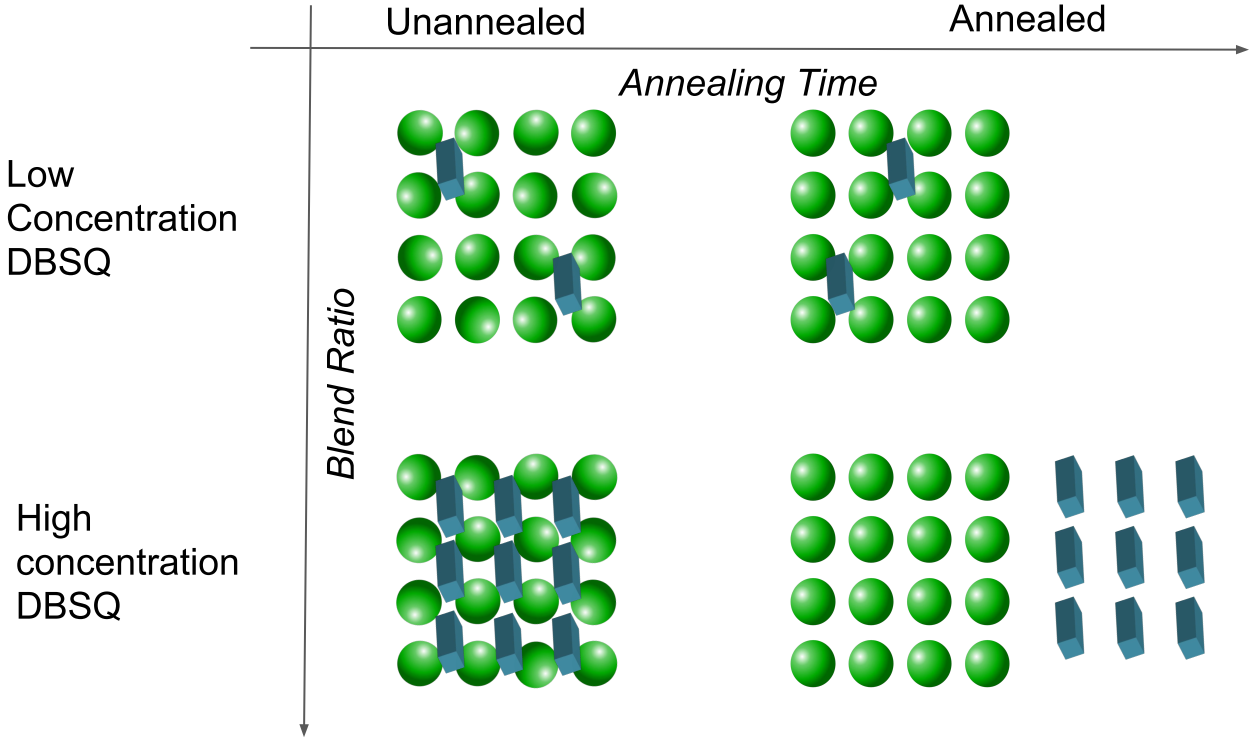


fewer pathways for current b/t acceptors



Squaraines aggregated but still not many pathways for current

Models of Molecule Intermixing



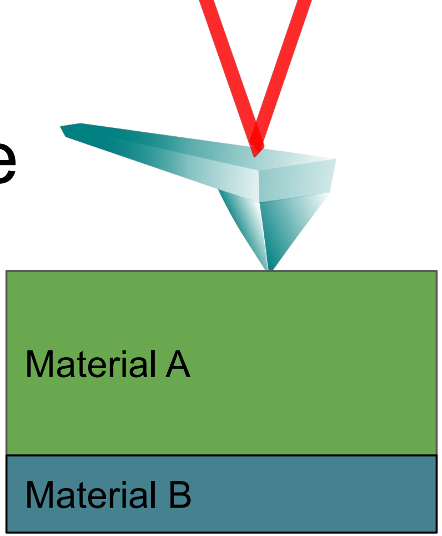
Thesis Goals:

1. *Understand the changes in morphology due to changes in blend of PCBM:DBSQ(OH)*$_2$*.*
2. *Understand the impact of annealing on morphology*

Oversimplified: One Model of Volume

$$\left(\frac{m_{PCBM}}{\rho_{PCBM}}\right) + \left(\frac{m_{DBSQ}}{\rho_{DBSQ}}\right) = V_{film}$$

- Assumes that the total volume of the materials is the sum of the individual volumes
- If the total volume (based on the densities of $DBSQ(OH)_2$ and PCBM) is not the sum of the volumes of these two materials -> we have evidence of intermixing and morphology changes

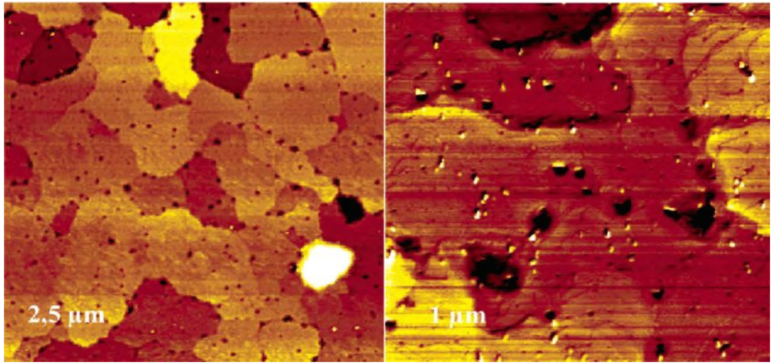


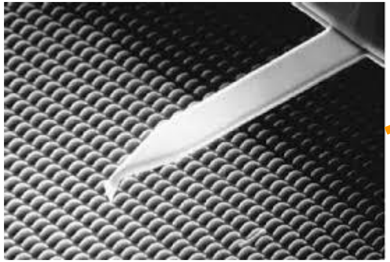
How we Measure Volume and Density

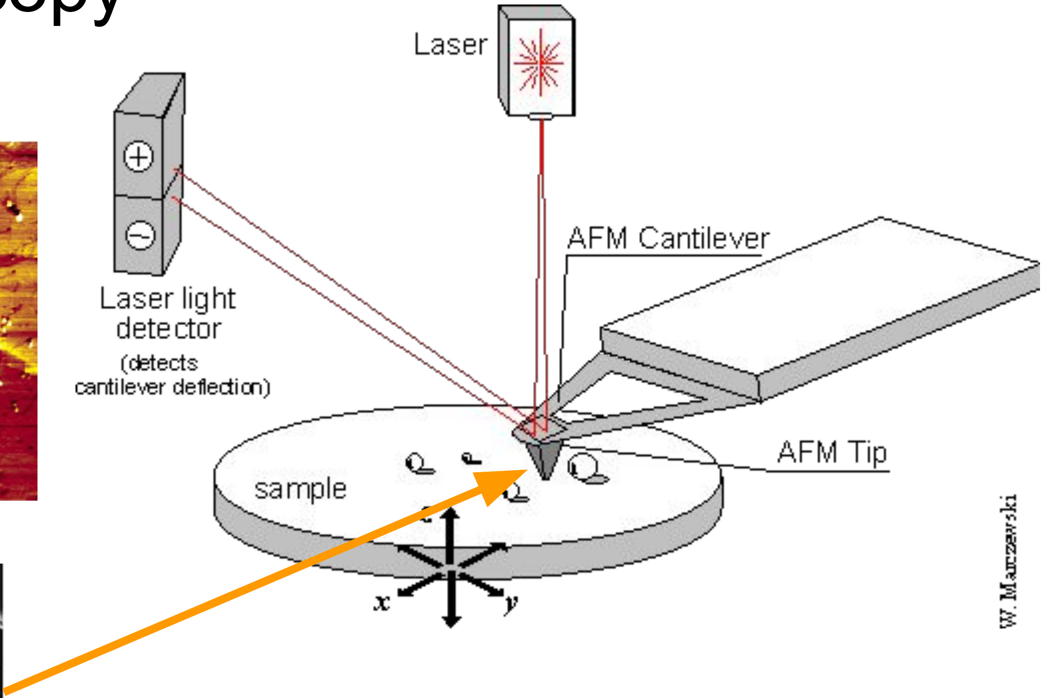
Height Measurements	Area Measurements	Mass Measurements
Atomic Force Microscopy	ImageJ Analysis of Photographs	Re-dissolved films/ Spectroscopy

100% PCBM - serve as reference point, as this is ϱ known from XRD

Atomic Force Microscopy

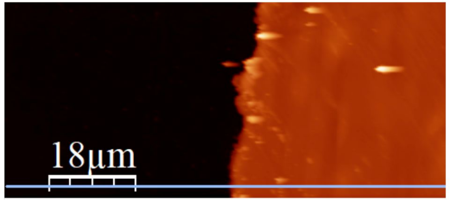


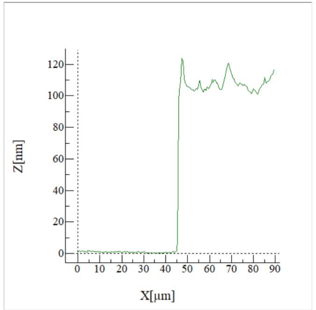


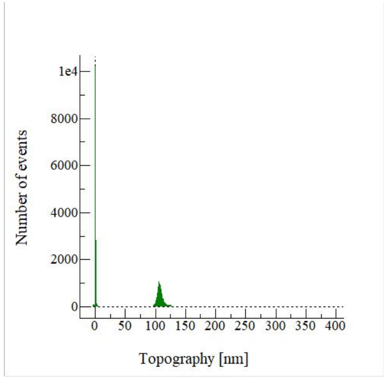


Height Measurements

- Atomic Force Microscopy was used to analyze the sample heights
- Images were analyzed in WSxM
  - 4 samples
  - 77 Images analyzed (~10 unannealed and ~10 annealed for each sample)
  - 2 trace images and 2 retrace images of each image were used
  - ~308 total unique data points measured this semester!
  - Including previous data, 11 samples were used ~ 880 total data points!

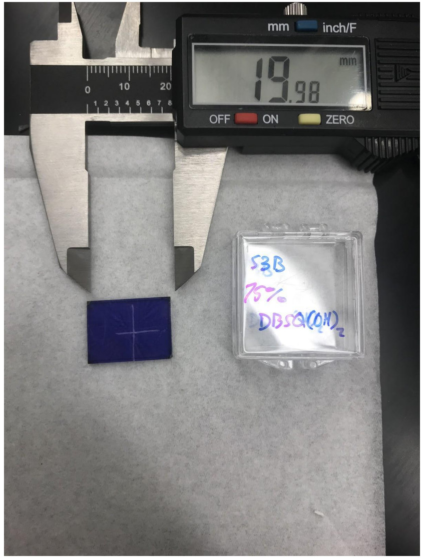


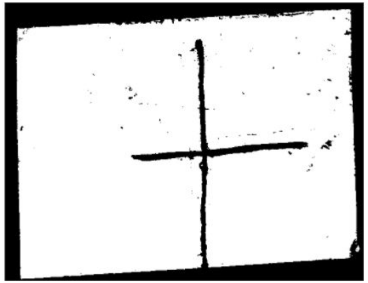




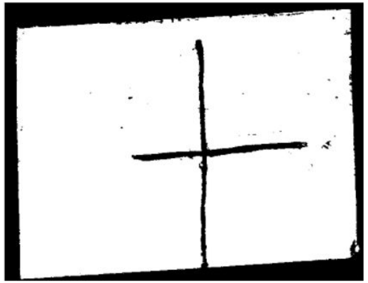
Area Measurements

- Image J was used
- Photographic images were converted into 32-bit binary images.
- Thresholding technique developed to determine the area and uncertainty

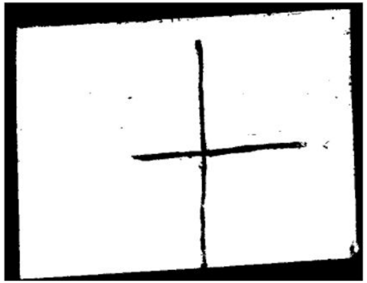




1



2

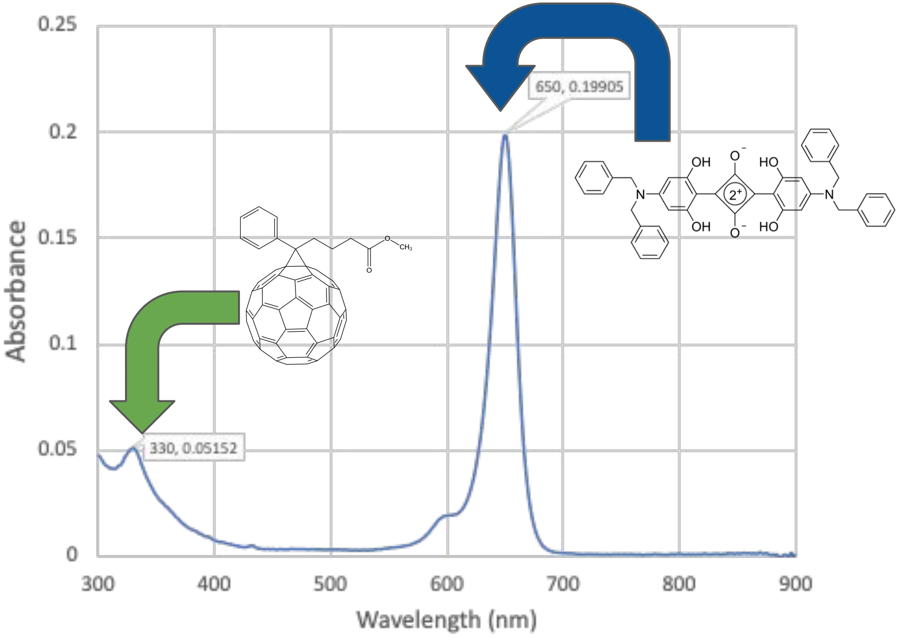


3

Mass Measurements

- Films were Re-Dissolved by collaborator at RIT, Tyler Wiegand
- Beer's law was used to determine solution concentrations of $DBSQ(OH)_2$ (max absorbance at 650 nm) and PCBM (max absorbance at 330 nm)
- Extinction coefficients well known at these wavelengths

$$A=\varepsilon cl$$



Mass Measurements (continued)

- Evidence of systematic error- unlikely that 12-15% of 100%DBSQ blend is contaminated with PCBM
- This systematic error is most likely due to background interference from PCBM spectrum at 650 nm and from $DBSQ(OH)_2$ spectrum at 330 nm.

Target SQ wt.%	Solution SQ wt.%	DBSQ(OH)2 Mass (g)	PCBM Mass (g)	Actual SQ wt.%
0%	0%	1.44E-06	7.32E-05	1.93%
0%	0%	4.71E-06	7.24E-05	6.11%
25%	24.37%	1.72E-05	5.56E-05	23.66%
40%	39.72%	3.45E-05	5.61E-05	38.12%
40%	39.72%	3.84E-05	5.63E-05	40.59%
50%	49.11%	3.26E-05	3.89E-05	45.60%
50%	49.11%	3.28E-05	3.62E-05	47.59%
75%	74.08%	5.24E-05	2.14E-05	71.00%
75%	74.08%	6.42E-05	2.74E-05	70.11%
100%	100%	7.54E-05	1.30E-05	85.29%
100%	100%	7.88E-05	1.05E-05	88.22%

What can we learn from this?

1. Densities

2. Impact of Annealing

3. Validity of the Oversimplified Model

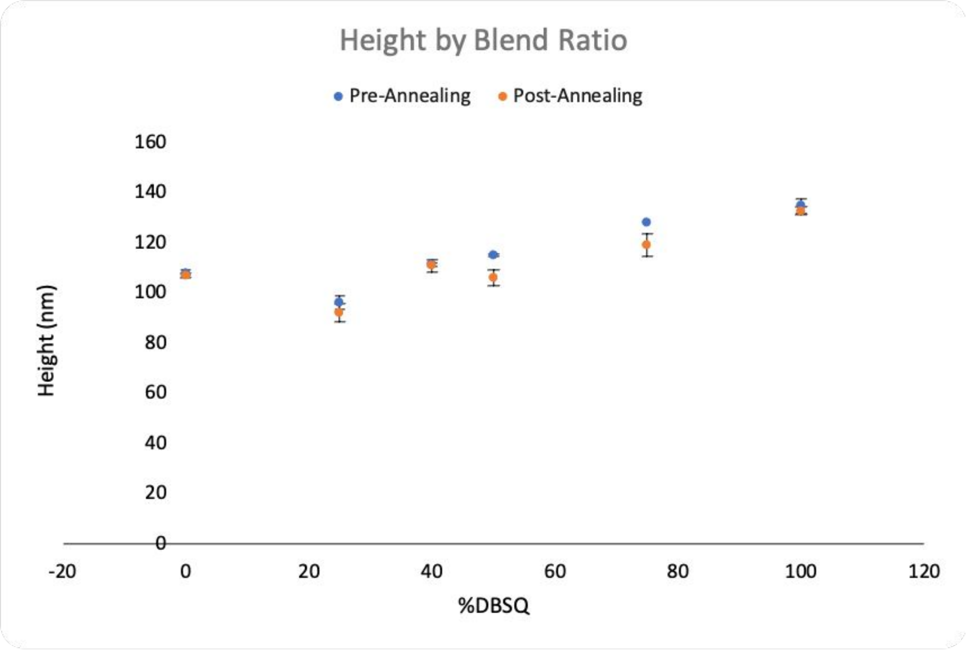
Inconsistent Densities of PCBM

- Measured density of PCBM is different from the density measured using X-ray Diffraction (1.631g/cm^3) and the amorphous density (1.5g/cm^3)
- This is most likely due to systematic error

	Density	Uncertainty	XRD Density
PCBM Unannealed	2.90E+00	1.27E-03	1.5
PCBM annealed	2.90E+00	1.36E-03	1.631

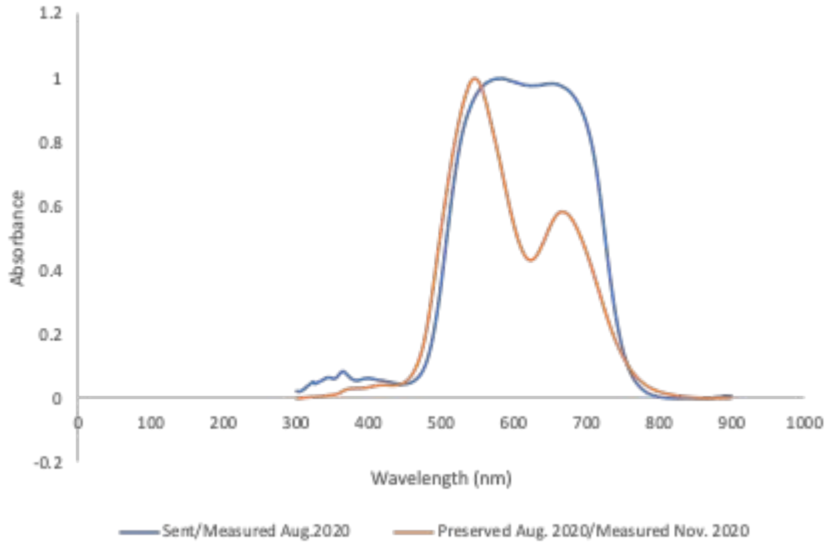
Impact of Annealing

- Annealing also has an impact on heights, which play into our density calculations
- Did not see differences in 0% or 100% $DBSQ(OH)_2$ blends
- Heights of intermediate blend ratios are clearly changing

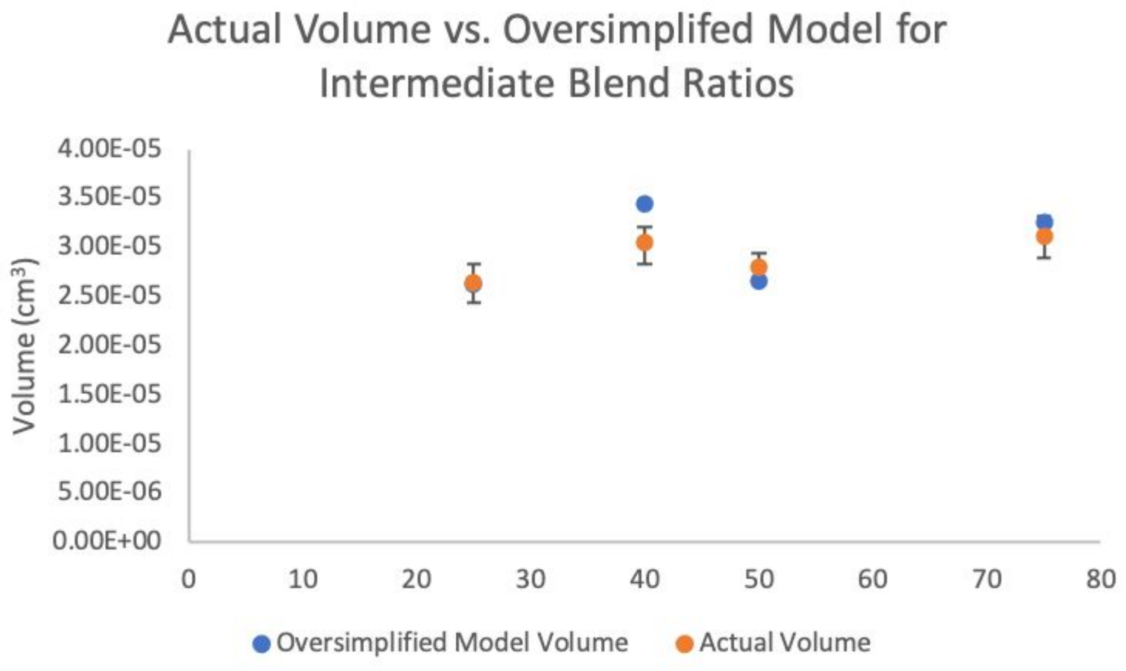


Film Changes over Time

- Also impacting heights is the changes in the film over time
- Intermediate blend ratios were measured during the summer by professor Burson, while I measured the 100% $DBSQ(OH)_2$ and 0% $DBSQ(OH)_2$ blend ratios
- This data on a sample film from the same batch at RIT shows double-hump character (indication of aggregates formed in $DBSQ(OH)_2$



Assessing the Validity of the Oversimplified Model



$$\left(\frac{m_{PCBM}}{\rho_{PCBM}}\right) + \left(\frac{m_{DBSQ}}{\rho_{DBSQ}}\right) = V_{film}$$

Conclusions

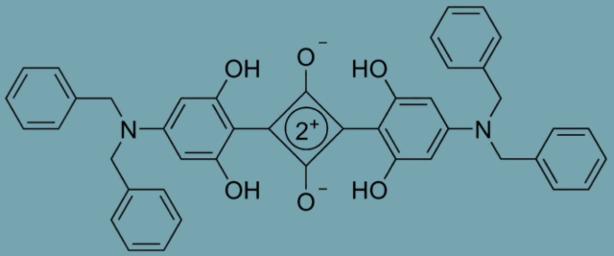
Preliminary results suggest deviation from the oversimplified model, which points to aggregation of DBSQ(OH)$_2$ as the cause of a deviation in density.

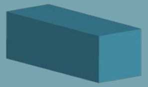
Clear that there is an impact of annealing on heights of intermediate blend ratios, which suggests that annealing increases aggregation in these devices

Future Improvements

- Reduce impact of systematic error in masses
  - We are in discussions with our collaborators now
- Reduce impact of random error in the area analysis
  - Using a well-defined procedure to produce high quality thresholds
  - Making sure that images are free of defects, including glare and false scratches
  - Potentially developing a new procedure entirely for ImageJ, or using a different thresholding tool
- Reduce impact of films changing over time (or quantify this impact)
  - The process of AFM analysis takes quite a bit of time, which provides an issue here

Donor: DBSQ

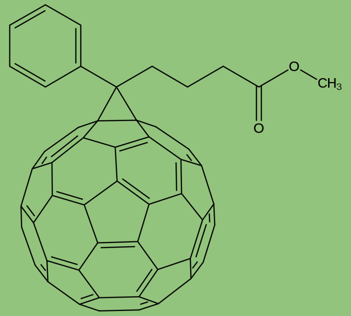


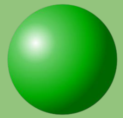


Benefits:

- Efficient photovoltaic material
- Reactivity to 600nm-850nm light sources (visible and infrared)
- Easy to produce
- Aggregates = good electron transport
- Higher functionality than other commonly used donor molecules

Acceptor: PCBM





Benefits:

- Well-studied acceptor
- Commercially available
- Good HOMO alignment with DBSQ -> electron transport
- Side chain enhances solubility in solution-based OSCs
- C60 allows for high open circuit voltage -> improved electron transfer with DBSQ